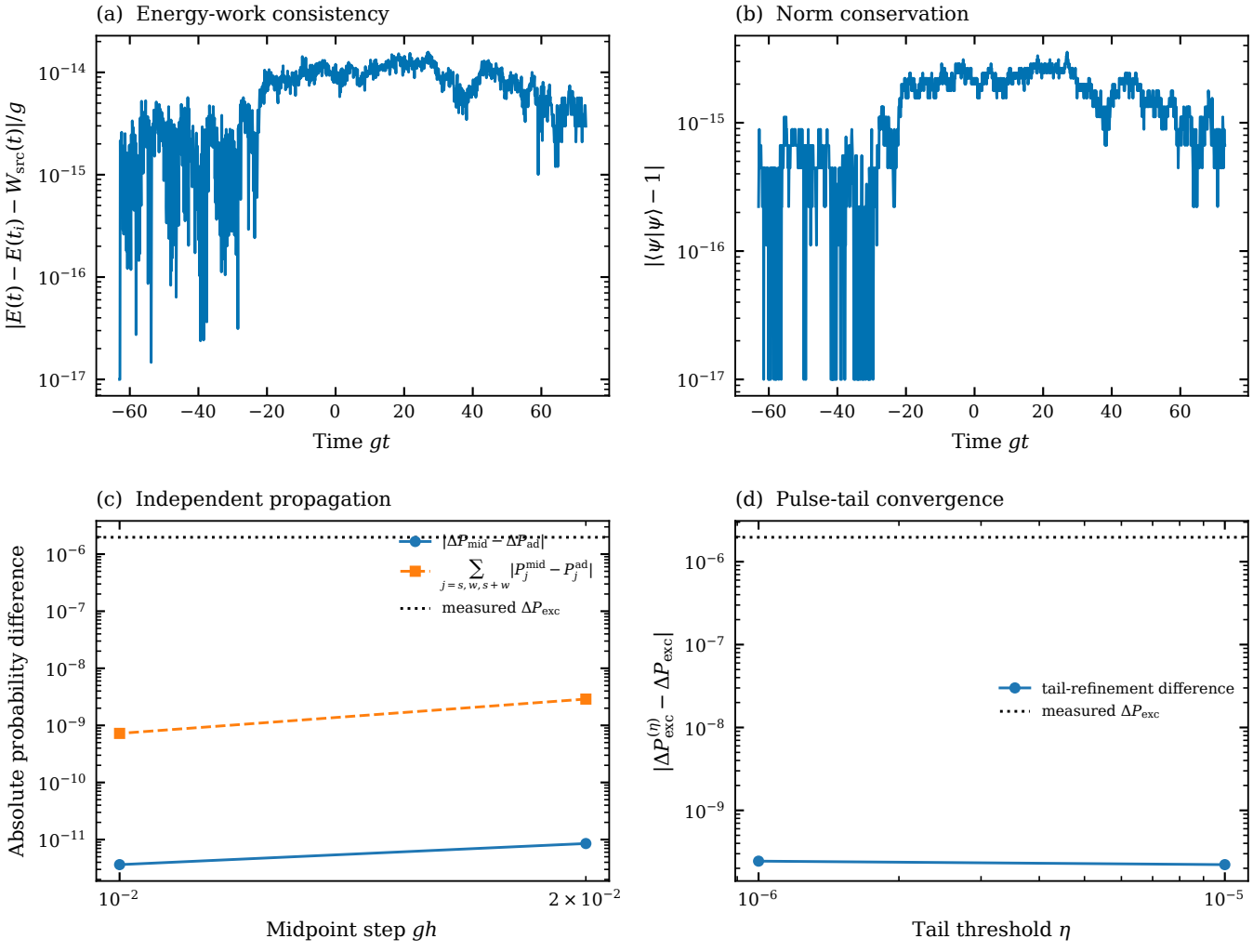

(a) Energy-work consistency
$|E(t) - E(t_i) - W_{\mathrm{src}}(t)|/g$
10^{-14}
10^{-15}
10^{-16}
10^{-17}
−60
−40
−20
0
20
40
60
Time gt
(b) Norm conservation
$|\langle\psi|\psi\rangle - 1|$
10^{-15}
10^{-16}
10^{-17}
−60
−40
−20
0
20
40
60
Time gt
(c) Independent propagation
Absolute probability difference
$|\Delta P_{\mathrm{mid}} - \Delta P_{\mathrm{ad}}|$
$\sum_{j=s,w,s+w} |P_j^{\mathrm{mid}} - P_j^{\mathrm{ad}}|$
measured ΔP_{exc}
10^{-6}
10^{-7}
10^{-8}
10^{-9}
10^{-10}
10^{-11}
10^{-2}
2×10^{-2}
Midpoint step gh
(d) Pulse-tail convergence
$|\Delta P_{\mathrm{exc}}^{(\eta)} - \Delta P_{\mathrm{exc}}|$
tail-refinement difference
measured ΔP_{exc}
10^{-6}
10^{-7}
10^{-8}
10^{-9}
10^{-6}
10^{-5}
Tail threshold η